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Synopsis

On

**“AI - BASED FACIAL RECOGNITION ATTENDANCE
MANAGEMENT SYSTEM”**

Submitted in partial fulfilment for the requirement of

Mini Project [BIS586]

In

INFORMATION SCIENCE & ENGINEERING

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BANGALORE COLLEGE OF ENGINEERING AND TECHNOLOGY



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CERTIFICATE

This is to certify that the mini project work , entitled “**AI-BASED FACIAL RECOGNITION ATTENDANCE MANAGEMENT SYSTEM**” is a Bonafide work carried out by **SWETHA , SAGAR SAWARN , YASHWARDHAN JHA** in partial fulfilment for the award of degree of Bachelor of Engineering in Information Science and Engineering during the academic year 2025-26.It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the Report .The mini project has been approved as it satisfies the academic requirements.

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We, Swetha, Sagar Sawarn ,Yashwardhan Jha hereby declare that the entire work titled “**AI-BASED FACIAL RECOGNITION ATTENDANCE MANAGEMENT SYSTEM**” embodied in this mini project report has been carried out by me during the 5th semester of BE(ISE) degree at BCET in Visvesvaraya Technological University, Belagavi under the esteemed guidance of **Mrs. Swathi Priya**, Assistant Prof, Dept. of ISE, BCET. The work embodied in this dissertation work is original and it has not been submitted in part or full for any other degree in any University. Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University. We also declare that any Intellectual Property Rights generated out of this project carried out at BCET will be the property of BCET, Bangalore and we will be one of the authors of the same.

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ABSTRACT

In colleges, universities, organizations, schools, and offices, taking attendance is one of the most important tasks that must be done on a daily basis. The majority of the time, it is done manually, such as by calling by name or by roll number. The main goal of this project is to create a Face Recognition-based attendance system that will turn this manual process into an automated one. This project meets the requirements for bringing modernization to the way attendance is handled, as well as the criteria for time management. This device is installed in the classroom, where and student's information, such as name, roll number, class, sec, and photographs, is trained. The images are extracted using Open CV. Before the start of the corresponding class, the student can approach the machine, which will begin taking pictures and comparing them to the qualified dataset. Logitech C270 web camera and NVIDIA Jetson Nano Developer kit were used in this project as the camera and processing board. The image is processed as follows: first, faces are identified using a Haar cascade classifier, then faces are recognized using the LBPH (Local Binary Pattern Histogram) Algorithm. Histogram data is checked against an established dataset and the device automatically labels attendance. An Excel sheet is developed, and it is updated every hour with the information from the respective class instructor.

TABLE OF CONTENTS

Abstract.....	vi
List of Figures.....	viii
List of Tables.....	viii

CHAPTER 1 - INTRODUCTION	03-05
CHAPTER 2 - LITERATURE REVIEW	06-08
CHAPTER 3 - MODEL IMPLEMENTATION AND ANALYSIS.....	09-12
CHAPTER 4 - CODE IMPLEMENTATION.....	13-15
CHAPTER 6 - PERFORMANCE ANALYSIS.....	16
CONCLUSION.....	17
REFERENCES.....	18

LIST OF FIGURES

Figure 1.1 Project outline	06
Figure 2.1 A diagram showing the steps in digital image	08
Figure 3.1 Model Implement	10
Figure 3.2 Installing OpenCV	12
Figure 4.4 Output Image 1	20
Figure 4.4 Output Image 2	20
Figure 4.4 Output Image 3	21
Figure 4.4 Output Image 4	21
Figure 6.1 Flow chart	22

LIST OF TABLES

TABLE	Page No.
2.1 Advantages and Disadvantages of Difference Biometric.	06
2.2 Advantages and Disadvantages of Face Detection Method.	08

CHAPTER - 1

INTRODUCTION

1.1 Project Objective

Attendance is prime important for both the teacher and student of an educational organization. So it is very important to keep record of the attendance. The problem arises when we think about the traditional process of taking attendance in class room.

Calling name or roll number of the student for attendance is not only a problem of time consumption but also it needs energy. So an automatic attendance system can solve all above problems.

There are some automatic attendances making system which are currently used by much institution. One of such system is biometric technique and RFID system. Although it is automatic and a step ahead of traditional method it fails to meet the time constraint. The student has to wait in queue for giving attendance, which is time taking.

This project introduces an involuntary attendance marking system, devoid of any kind of interference with the normal teaching procedure. The system can be also implemented during exam sessions or in other teaching activities where attendance is highly essential.

This system eliminates classical student identification such as calling name of the student, or checking respective identification cards of the student, which can not only interfere with the ongoing teaching process, but also can be stressful for students during examination sessions. In addition, the students have to register in the database to be recognized. The enrolment can be done on the spot through the User friendly interface.

1.2 Background

Face recognition is crucial in daily life in order to identify family, friends or someone we are familiar with. We might not perceive that several steps have actually taken in order to identify human faces. Human intelligence allows us to receive information and interpret the information in the recognition process.

We receive information through the image projected into our eyes, by specifically retina in the form of light. Light is a form of electromagnetic waves which are radiated from a source onto an object and projected to human vision.

Robinson-Riegler, G., & Robinson-Riegler, B. (2008) mentioned that after visual processing done by the human visual system, we actually classify shape, size, contour and the texture of the object i

The information will be compared to other representations of objects or face that exist in our memory to recognize. In fact, it is a hard challenge to build an automated system to have the same capability as a human to recognize faces.

Attendance management is crucial in educational institutions and workplaces. Conventional methods, such as manual roll calls or RFID systems, are often inefficient and error-prone. The emergence of AI-based facial recognition provides an opportunity to automate attendance, ensuring **accuracy, rapid processing, and secure identification**. The system captures facial images through a camera, processes them using facial recognition algorithms, and updates a database in real-time. This integration reduces teacher/staff workload, streamlines administrative tasks, and enables data-driven decision-making.

1.3 Problem Statement

Traditional student attendance marking technique is often facing a lot of trouble. The face recognition student attendance system emphasizes its simplicity by eliminating classical student attendance marking technique such as calling student names or checking respective identification cards.

There are not only disturbing the teaching process but also causes distraction for students during exam sessions. Apart from calling names, attendance sheet is passed around the classroom during the lecture sessions. The lecture class especially the class with a large number of students might find it difficult to have the attendance sheet being passed around the class.

Thus, face recognition attendance system is proposed in order to replace the manual signing of the presence of students which are burdensome and causes students get distracted in order to sign for their attendance. Furthermore, the face recognition based automated student attendance system able to overcome the problem of fraudulent approach and lecturers does not have to count the number of students several times to ensure the presence of the students.

The paper proposed by Zhao, W et al. (2003) has listed the difficulties of facial identification. One of the difficulties of facial identification is the identification between known and unknown images. In addition, paper proposed by Pooja G.R et al. (2010) found out that the training process for face recognition student attendance system is slow and time-consuming . In addition, the paper proposed by Priyanka Wagh et al. (2015) mentioned that different lighting and head poses are often the problems that could degrade the performance of face recognition based student attendance system Hence, there is a need to develop a real time operating student attendance system which means the identification process must be done within defined time constraints to prevent omission.

The extracted features from facial images which represent the identity of the students have to be consistent towards a change in background, illumination, pose and expression. High accuracy and fast computation time will be 5 the evaluation points of the performance.

1.4 Aims and Objectives

The objective of this project is to develop face recognition attendance system. Expected achievements in order to Fullfill the objectives are:

- To detect the face segment from the video frame.
- To extract the useful features from the face detected.
- To classify the features in order to recognize the face detected.
- To record the attendance of the identified student.
- Automate attendance marking using facial recognition.
- Reduce time and administrative effort for attendance management.
- Enhance accuracy and prevent proxy or fraudulent attendance.
- Generate daily, weekly, and monthly attendance reports.
- Store attendance data securely in a database (e.g., **SQL**, **MongoDB**, or **Redis**).
- Provide realtime notifications and analytics for students and administrators.

1.5 Flowchart

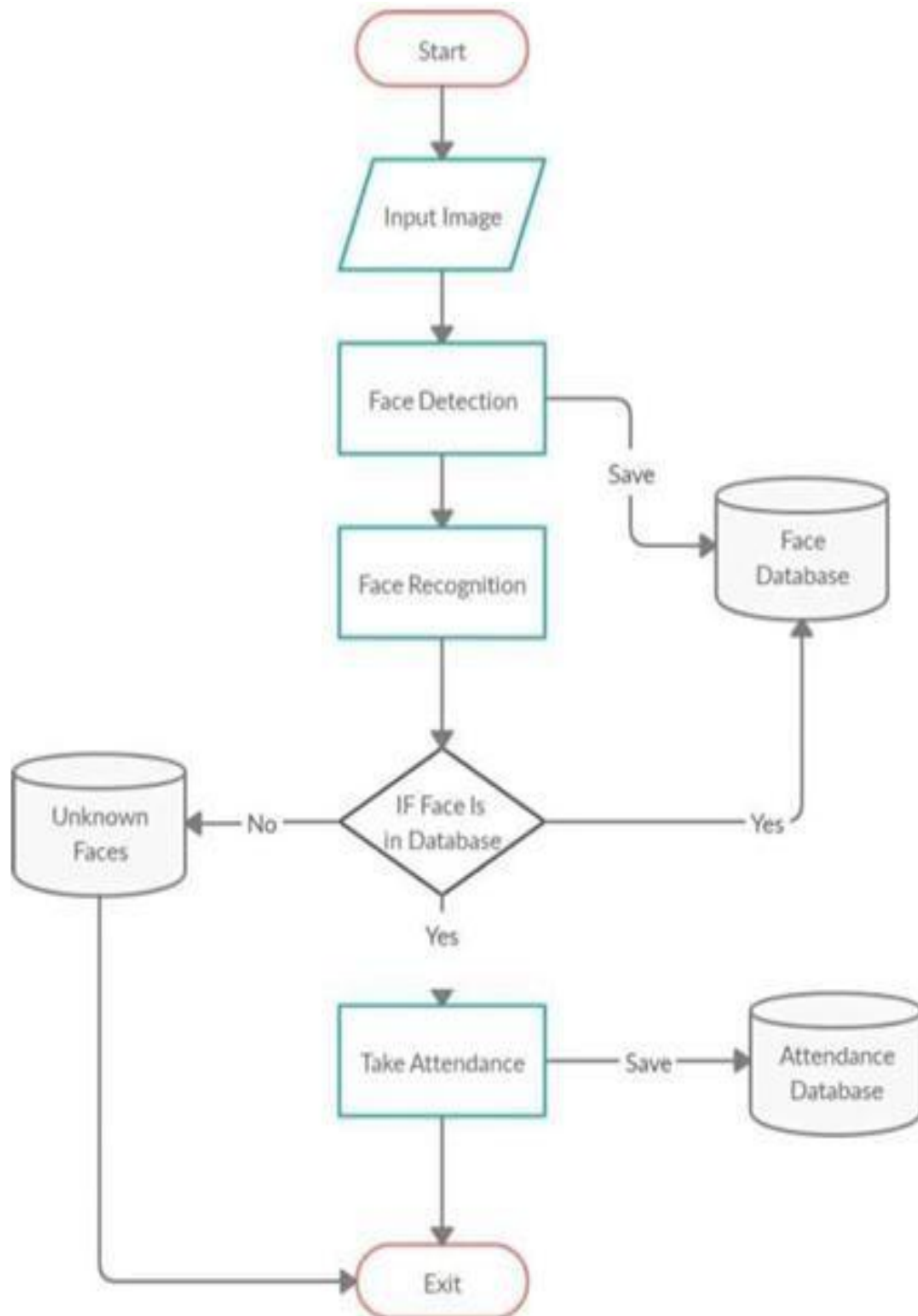


Figure 1.1 project outline

CHAPTER - 2

LITERATURE REVIEW

2.1 Student Attendance System

Arun Katara et al. (2017) mentioned disadvantages of RFID (Radio Frequency Identification) card system, fingerprint system and iris recognition system. RFID card system is implemented due to its simplicity. However, the user tends to help their friends to check in as long as they have their friend's ID card. The fingerprint system is indeed effective but not efficient because it takes time for the verification process so the user has to line up and perform the verification one by one. However for face recognition, the human face is always exposed and contain less information compared to iris. Iris recognition system which contains more detail might invade the privacy of the user. Voice recognition is available, but it is less accurate compared to other methods. Hence, face recognition system is suggested to be implemented in the student attendance system.

System Type	Advantages	Disadvantages
RFID card system	Simple	Fraudulent usage
Fingerprint system	Accurate	Time consuming
Voice recognition system		Less accuracy when compared to others
Iris recognition system	Accurate	Privacy invasion

Table 2.1: Advantages & Disadvantages of Different Biometric System

2.2 Digital Image Processing

Digital Image Processing is the processing of images which are digital in nature by a digital computer. Digital image processing techniques are motivated by three major applications mainly:

- Improvement of pictorial information for human perception
- Image processing for autonomous machine application
- Efficient storage and transmission.

2.3 Image Representation in a Digital Computer

An image is a 2-Dimensional light intensity function

$$f(x, y) = r(x, y) \times i(x, y) - (2.0)$$

A digital image $f(x, y)$ is discretized both in spatial co-ordinates by grids and in brightness by quantization.

Effectively, the image can be represented as a matrix whose row, column indices specify a point in the image and the element value identifies grey level value at that point. These elements are referred to as pixels or pels.

Typically following image processing applications, the image size which is used is **256 × 256**, elements, **640 × 480** pels or **1024 × 1024** pixels. Quantization of these matrix pixels is done at 8 bits for black and white images and 24 bits for coloured images (because of the three colour planes Red, Green and Blue each at 8 bits)

2.4 Steps in Digital Image Processing

Digital image processing involves the following basic tasks:

- Image Acquisition - An imaging sensor and the capability to digitize the signal produced by the sensor.
- Preprocessing – Enhances the image quality, filtering, contrast enhancement etc.
- Segmentation – Partitions an input image into constituent parts of objects.
- Description/feature Selection – extracts the description of image objects suitable for further computer processing.
- Recognition and Interpretation – Assigning a label to the object based on the information provided by its descriptor. Interpretation assigns meaning to a set of labelled objects.
- Knowledge Base – This helps for efficient processing as well as inter module cooperation.

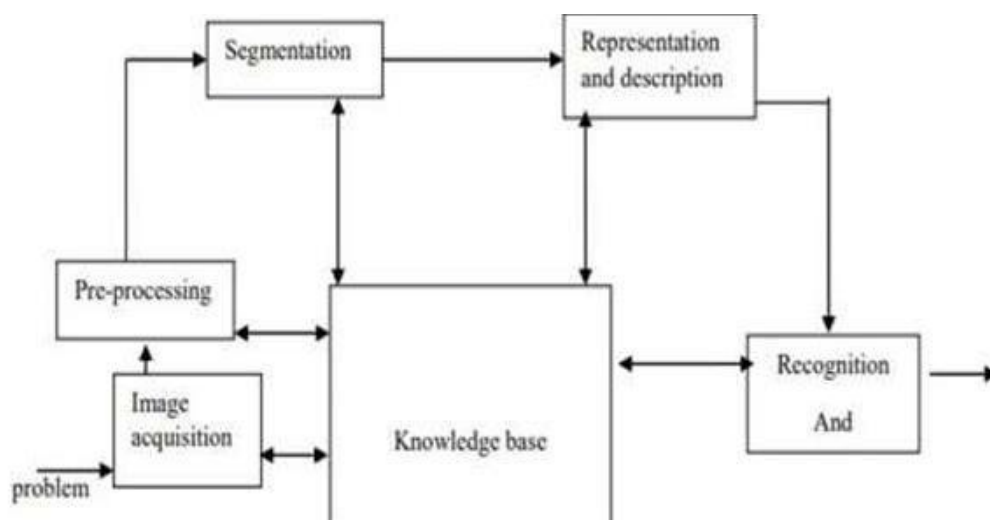


fig 2.1: A diagram showing the steps in digital image processing

2.5 Definition of Terms and History

Face Detection

Face detection is the process of identifying and locating all the present faces in a single image or video regardless of their position, scale, orientation, age and expression. Furthermore, the detection should be irrespective of extraneous illumination conditions and the image and video content[5].

Face Recognition

Face Recognition is a visual pattern recognition problem, where the face, represented as a three dimensional object that is subject to varying illumination, 12 pose and other factors, needs to be identified based on acquired images[6]. Face Recognition is therefore simply the task of identifying an already detected face as a known or unknown face and in more advanced cases telling exactly whose face it is[7].

Difference between Face Detection and Face Recognition

Face detection answers the question, Where is the face? It identifies an object as a “face” and locates it in the input image.

Face Recognition on the other hand answers the question who is this? Or whose face is it?

It decides if the detected face is someone .It can therefore be seen that face detections output (the detected face) is the input to the face recognizer and the face Recognition’s output is the final decision i.e. face known or face unknown.

Face Detection

Face Detection Method	Advantages	Disadvantages
Viola - Jones Algorithm	1)High detection speed 2)High accuracy	1) Long training time. 2) Limited head pose. 3)Not able to detect dark faces.
Local Binary Pattern Histogram	1)Simple computation. 2)High tolerance against monotonic illumination changes.	1)Only used for binary and grey images. 2)Overall performance is inaccurate compared to viola - jones algorithm.
Ada Boost Algorithm	Need not to have any prior knowledge about face structure	The result highly depends on the training data and affected by weak classifiers.
SMQT features and SNOW classifier method	1)Capable to deal with lighting problem in object detection. 2)Efficient in computation.	The region contain very similar to grey value regions will be misidentified as face

Table 2.2 Advantages and Disadvantages of Face Detection Methods

CHAPTER - 3

MODEL IMPLEMENTATION AND ANALYSIS

3.1 : INTRODUCTION

Face detection involves separating image windows into two classes; one containing faces (turning the background (clutter). It is difficult because although commonalities exist between faces, they can vary considerably in terms of age, skin color and facial expression. The problem is further complicated by differing lighting conditions, image qualities and geometries, as well as the possibility of partial occlusion and disguise. An ideal face detector would therefore be able to detect the presence of any face under any set of lighting conditions, upon any background. The face detection task can be broken down into two steps. The first step is a classification task that takes some arbitrary image as input and outputs a binary value of yes or no, indicating whether there are any faces present in the image. The second step is the face localization task that aims to take an image as input and output the location of any face or faces within that image as some bounding box with (x, y, width, height). After taking the picture the system will compare the equality of the pictures in its database and give the most related result.

We will use NVIDIA Jetson Nano Developer kit, Logitech C270 HD Webcam, open CV platform and will do the coding in python language.

3.2 Model Implementation

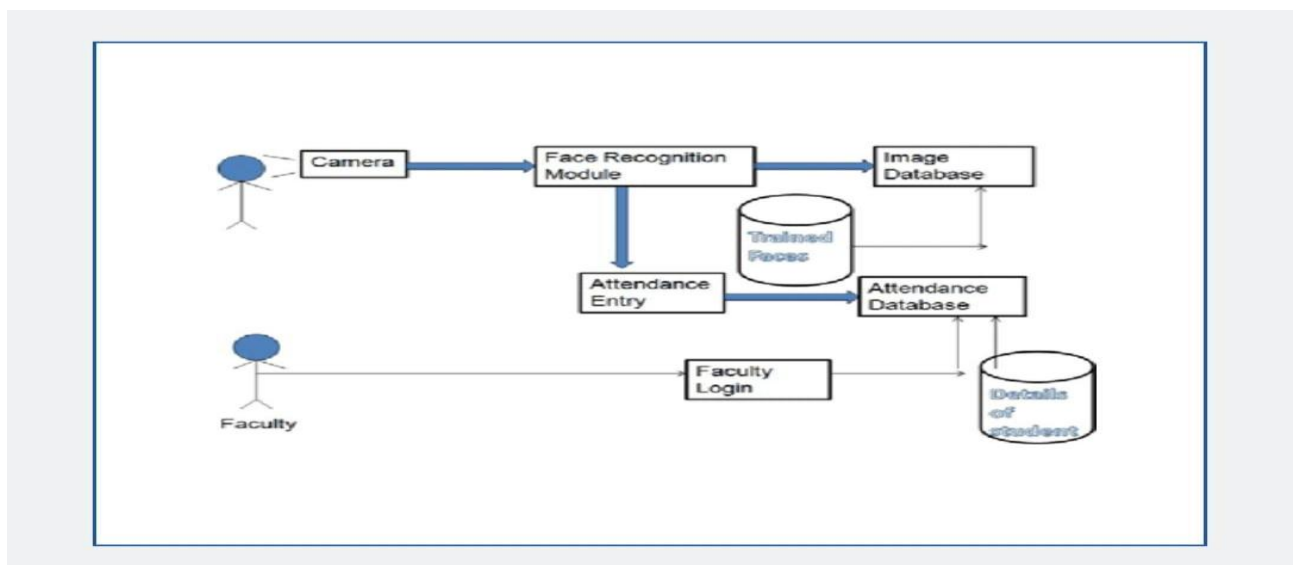


Figure 3.1 Model Implementation

The main components used in the implementation approach are open source computer vision library (OpenCV). One of OpenCV's goals is to provide a simple to use computer vision infrastructure that helps people build fairly sophisticated vision applications quickly. OpenCV library contains over 500 functions that span many areas in vision. The primary technology behind Face recognition is OpenCV. The user stands in front of the camera keeping a minimum distance of 50cm and his image is taken as an input. The frontal face is extracted from the image then converted to gray scale and stored.

The Principal component Analysis (PCA) algorithm is performed on the images and the eigen values are stored in an xml file. When a user requests for recognition the frontal face is extracted from the captured video frame through the camera. The eigen value is re-calculated for the test face and it is matched with the stored data for the closest neighbour.

3.3 Design Requirements

We used some tools to build the system. Without the help of these tools it would not be possible to make it done. Here we will discuss about the most important one.

3.3.1 Software Implementation

1)Open CV : We used OpenCV 3 dependency for python 3. OpenCV is library where there are lots of image processing functions are available. This is very useful library for image processing. Even one can get expected outcome without writing a single code. The library is cross-platform and free for use under the open-source BSD license. Example of some supported functions are given bellow:

- **Derivation:** Gradient/Laplacian computing, contours delimitation
- **Hough transforms:** lines, segments, circles, and geometrical shapes detection
- **Histograms:** computing, equalization, and object localization with back projection algorithm
- **Segmentation:** thresholding, distance transform, foreground/background detection, watershed segmentation
- **Filtering:** linear and nonlinear filters, morphological operations
- **Cascade detectors:** detection of face, eye, car plates
- **Interest points:** detection and matching
- **Video processing:** optical flow, background subtraction, camshaft (object tracking)
- **Photography:** panoramas realization, high definition imaging (HDR), image inpainting

So it was very important to install OpenCV. But installing OpenCV 3 is a complex process. How we did it is given below:

Step 1: Check Python installation

Open your Command Prompt or Terminal and type `--version`
If Python isn't installed,
download it from <https://www.python.org/downloads/>

Step 2: Install OpenCV

Run this command in your command prompt
`pip install opencv-python`
Also need advanced features like face recognition
or image manipulation
`pip install opencv-contrib-python`

Step 3: Install additional libraries

For facial recognition and attendance systems
`pip install numpy`
`pip install face-recognition`
`pip install dlib`
`pip install cmake`
`pip install pillow`
Tip: `pip install dlib-19.22.99-cp39-cp39-amd64.whl`
(Use the correct version for your Python
version and system)

Step 4: Verify installation

Check if OpenCV works correctly
`import cv2`
`print(cv2.__version__)`

Step 5: Optional – Create a Virtual Environment

To keep your project clean and organized,
`python -m venv attendance_env`

Fig 3.2 Installing open cv

We copied this script and placed it in a directory on our Raspberry Pi and saved it. Then through terminal we made this script executable and then ran it.

Sudo	chmod
755/myfile/pi/installopencv.bash	
Sudo /myfile/pi/installopencv.bash	

These are the command line we used.

2) Python IDE: There are lots of IDEs for python. Some of them are PyCharm, Thonny, Ninja, Spyder etc. Ninja and Spyder both are very excellent and free but we used Spyder as it feature- rich than ninja. Spyder is a little bit heavier than ninja but still much lighter than PyCharm. You can run them in pi and get GUI on your PC

```
1. sudo apt-get isntall spyder
```

through ssh-Y. We installed Spyder through the command line below.

CHAPTER - 4

CODE IMPLEMENTATION

4.1 Code Requirements

- Opencv(pip install opencv-python)
- Tkinter(Available in python)
- PIL (pip install Pillow)
- Pandas(pip install pandas)

4.2 What steps you have to follow??

- Download my Repository
- Create a TrainingImage folder in a project.
- Open a AMS_Run.py and change the all paths with your system path
- Run AMS_Run.py.

4.3 Project Structure

- After run you need to give your face data to system so enter your ID and name in box than click on Take Images button.
- It will collect 200 images of your faces, it save a images in TrainingImage folder
- After that we need to train a model(for train a model click on Train Image button.
- It will take 5-10 minutes for training(for 10 person data).
- After training click on Automatic Attendance ,it can fill attendace by your face using our trained model (model will save in TrainingImageLabel)
- it will create .csv file of attendance according to time & subject.
- You can store data in database (install wampserver),change the DB name according to your in AMS_Run.py.
- Manually Fill Attendace Button in UI is for fill a manually attendance (without facce recognition),it's also create a .csv and store in a database.

SAMPLE OUTPUTS

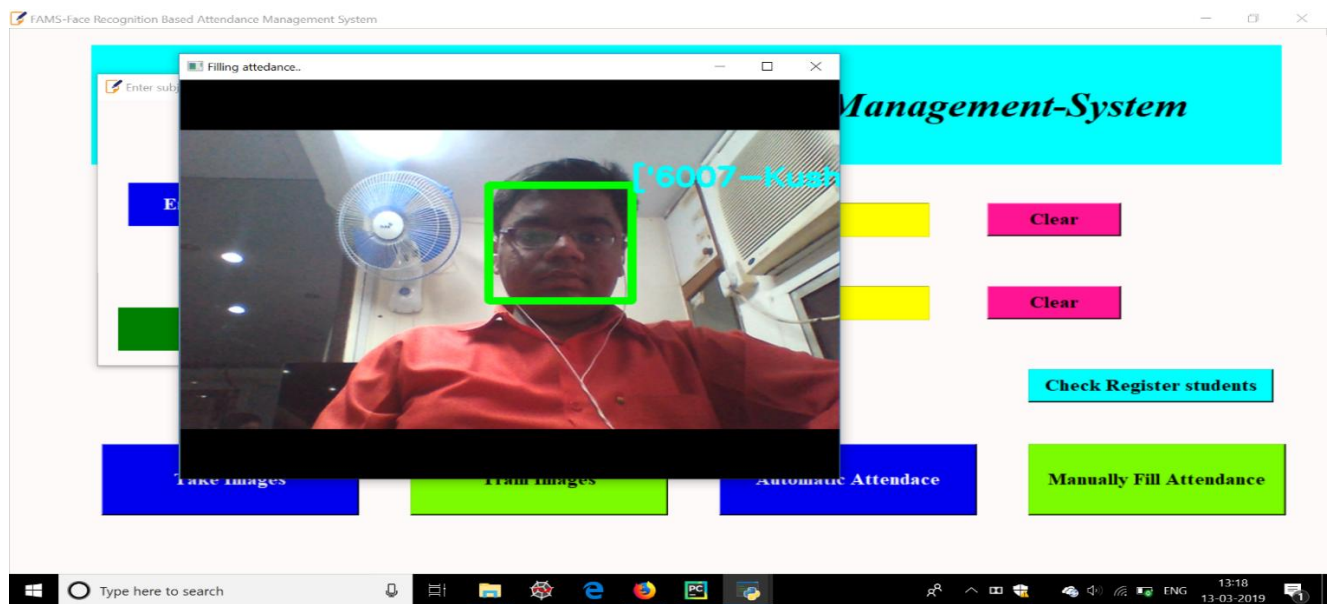


Figure 4.4 : Output Image 1

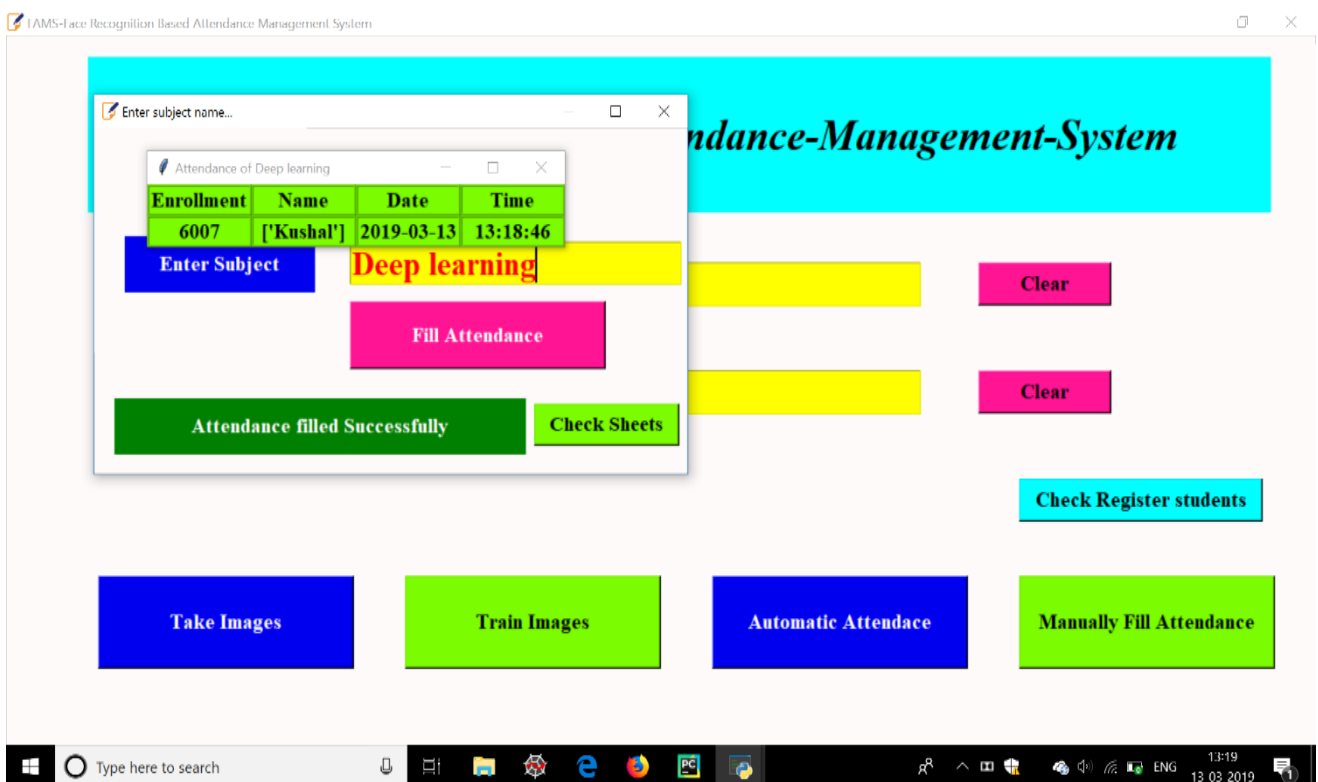


Figure 4.4 : Output Image 2

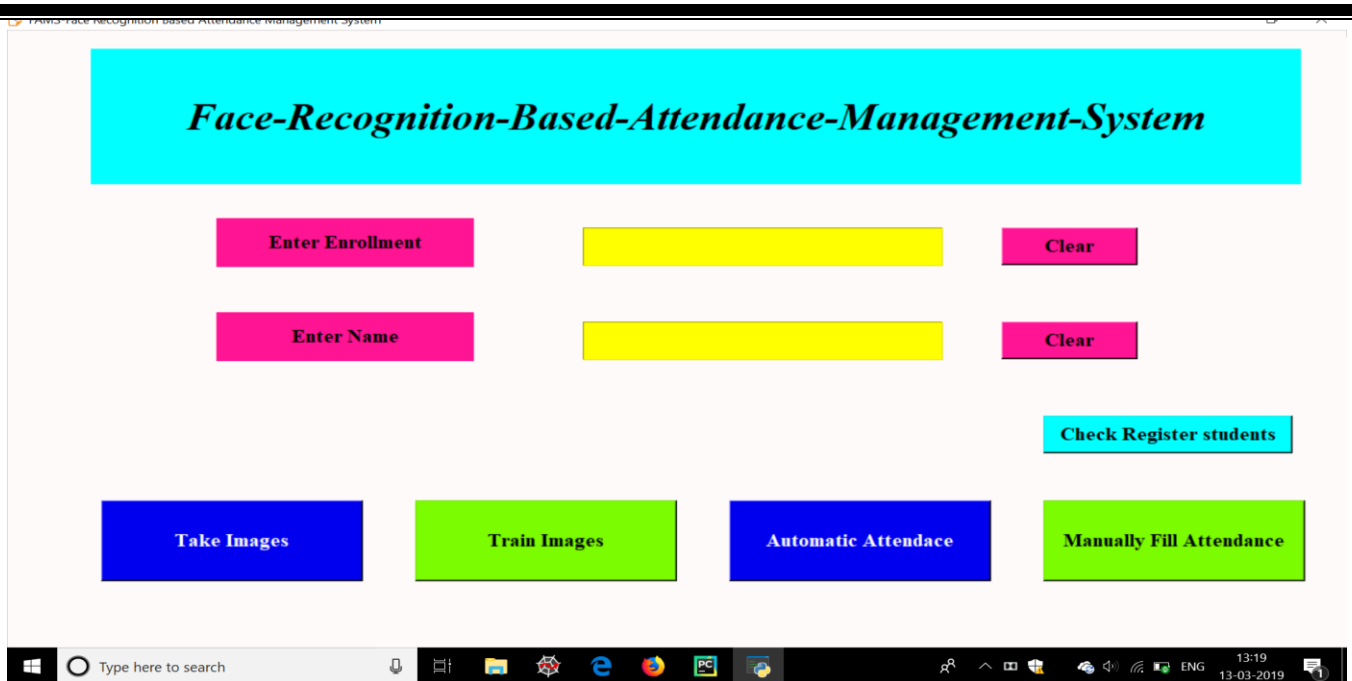


Figure 4.4 : Output Image 3

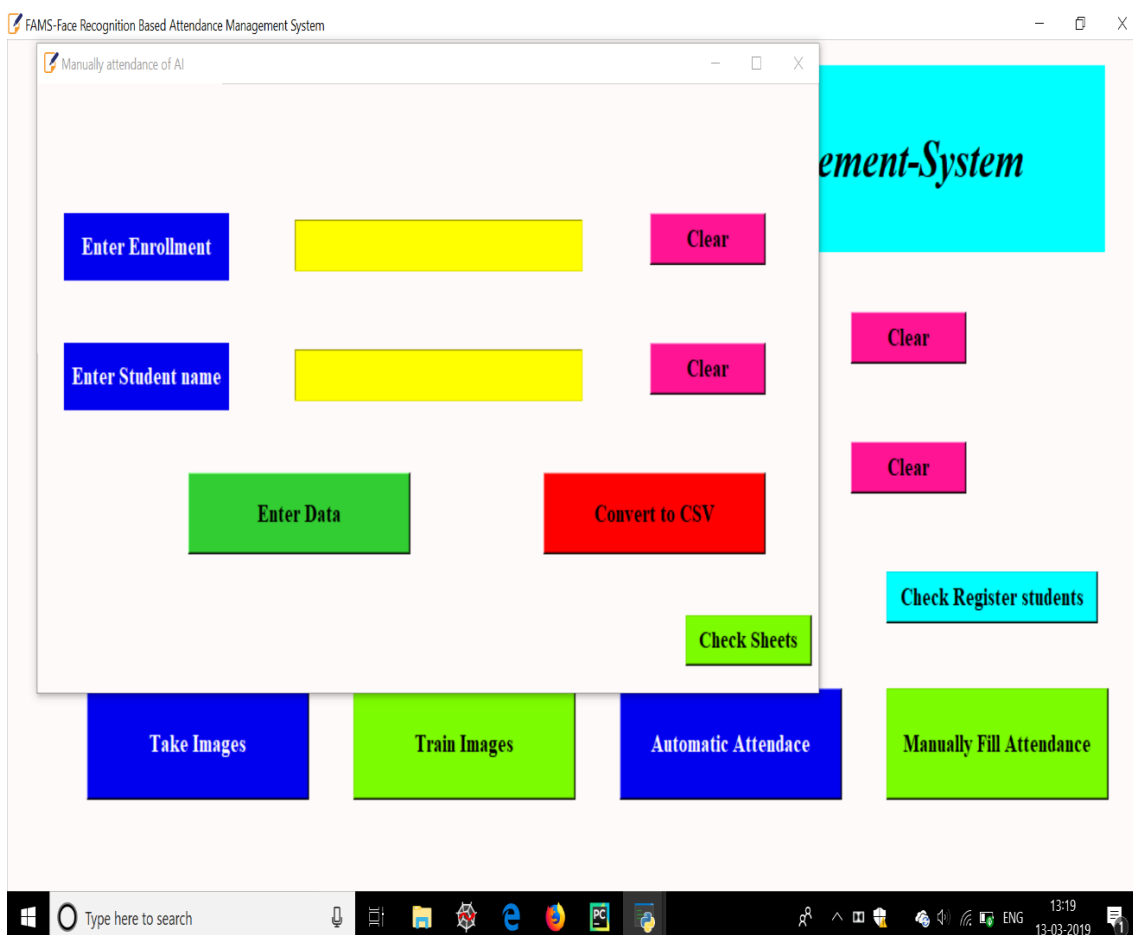


Figure 4.4 : Output Image 4

CHAPTER – 5

PERFORMANCE ANALYSIS

Analysis:

Here we consider one constant parameter intensity of light . we performed different experiments on different distance and different angles. we observed the confidence level at the different positions by gradually increasing the distance .we plotted the graph using the x and y coordinates by considering the x values as the confidence level or accuracy rate. and y values as the distance (cms).

Flow Chart:

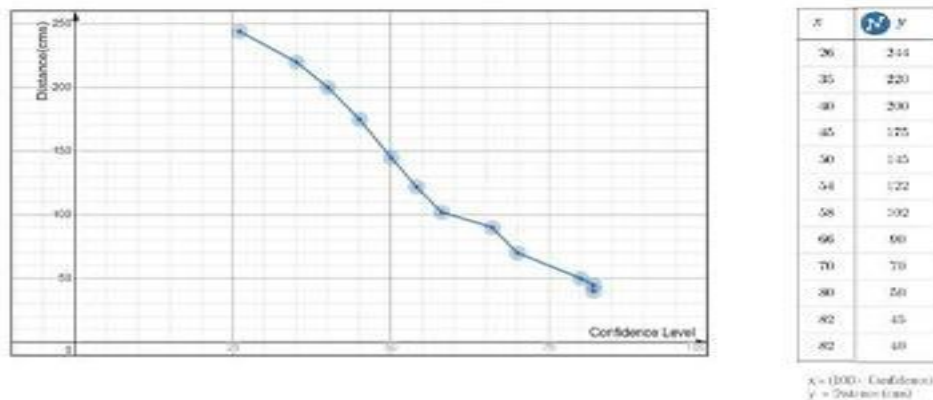


Figure 6.1 Flow chart

CONCLUSION

Face recognition systems are part of facial image processing applications and their significance as a research area are increasing recently. Implementations of system are crime prevention, video surveillance, person verification, and similar security activities. The face recognition system implementation can be part of Universities. Face Recognition Based Attendance System has been envisioned for the purpose of reducing the errors that occur in the traditional (manual) attendance taking system. The aim is to automate and make a system that is useful to the organization such as an institute. The efficient and accurate method of attendance in the office environment that can replace the old manual methods. This method is secure enough, reliable and available for use. Proposed algorithm is capable of detect multiple faces, and performance of system has acceptable good results.

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